

Optimization of Regularized Matrix Factorization for Movie Recommendation Systems

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Problem

- User-item rating matrices in recommender systems are extremely sparse.
- Predicting missing ratings is essential for personalized recommendations.
- We address this as a **matrix completion** and **optimization** problem.

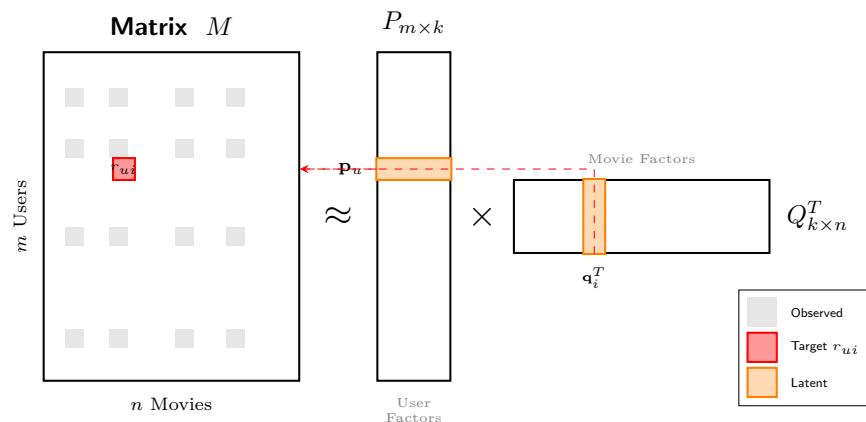
Goal: Learn a low-rank approximation that minimizes prediction error on unseen ratings.

Method

We model the rating matrix R using latent user and item factors:

$$\min_{U,V} \sum_{(i,j) \in \Omega} (R_{ij} - u_i v_j^T)^2 + \lambda(\|u_i\|^2 + \|v_j\|^2)$$

- Solver: **Stochastic Gradient Descent (SGD)** or **Alternating Least Squares (ALS)**.
- Bias terms: global, user, and item biases.
- Hyperparameters: latent dimension K , learning rate α , regularization λ .



Data & Experimental Setup

Dataset: MovieLens Latest Small.

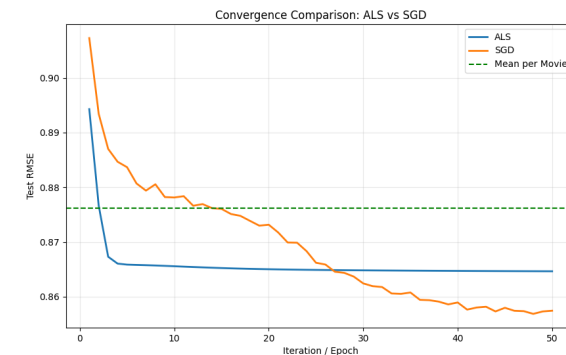
Contains $\sim 100K$ ratings, 600 users, 9K movies, split into 80-20 train-test sets, evaluated on RMSE. **Baselines:** Global mean predictor, Column-based predictor

Results & Analysis

- SGD converges steadily for all tested K values.
- Larger K improves accuracy but increases overfitting risk.
- Generalization gap highlights bias-variance trade-off.

Key Result:

SGD-based MF outperforms the baseline and achieves rational RMSE.



Conclusions

- Matrix Factorization is effective for sparse recommendation problems.
- SGD offers flexibility and fine-grained optimization control, ALS provides a strong comparison but is less scalable for dense tuning.
- **Future work:** implicit feedback, adaptive learning rates, hybrid models.